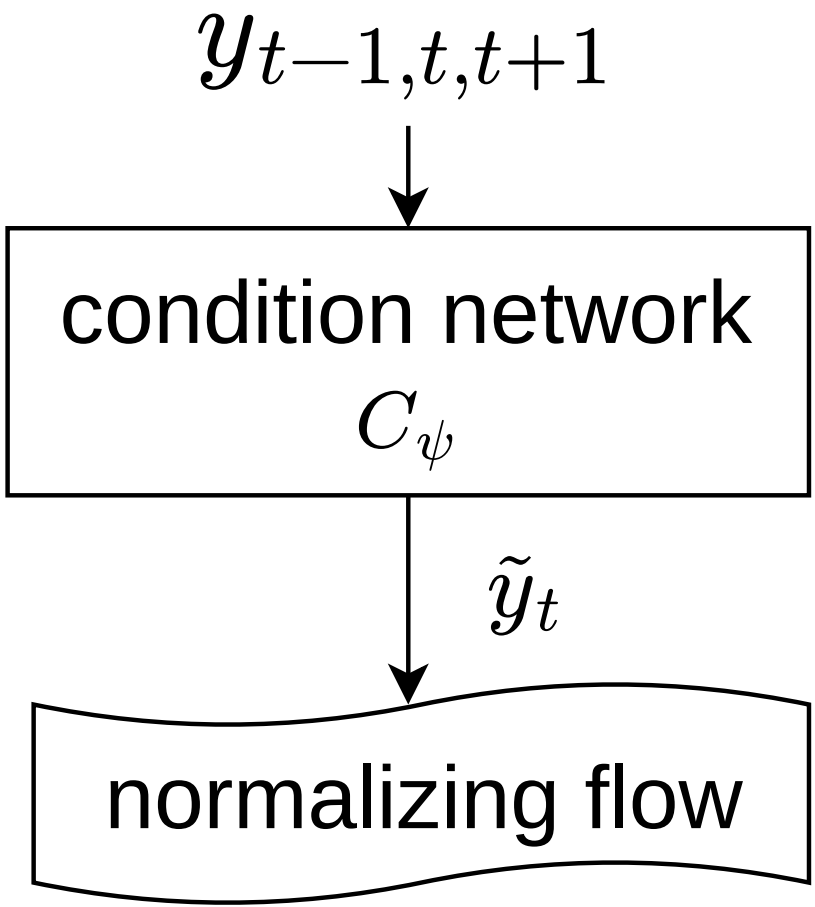

$$z_t = F z_{t-1} + G$$

$$G \sim N(0, \sigma I)$$